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UNIT FOUR Clinical Skills and Care

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CHAPTER 23 Nutrition

KEY TERMS

Absorption (ab-ZORB-shun)
Anabolism (a-NAB-uh-lizm)
Antioxidant (AN-ti-OX-i-dent)
Body mass index (BAH-dee MASS IN-deks)
Catabolism (kuh-TAB-uh-lizm)
Complete proteins (kom-PLEET PRO-teenz)
Complex carbohydrates (KOM-pleks KAR-boh-HYE-drayts)
Digestion (dye-JES-chun)
Essential amino acids (e-SEN-shul ah-MEE-noh ASS-ids)
Incomplete proteins (IN-kom-PLEET PRO-teenz)
Nitrogen balance (NYE-troh-jen BA-lanss)
Nonessential amino acids (NON-e-SEN-shul ah-MEE-noh ASS-ids)
Simple carbohydrates (SIM-puhl KAR-boh-HYE-drayts)

CHAPTER CONCEPTS

Growth and Development
Health Promotion
Nutrition

LEARNING OUTCOMES

1. Define various terms associated with nutrition.
2. Describe nutritional guidelines, including food label information, dietary guidelines for Americans, and My Plate recommendations.
3. Identify the six essential nutrients and the caloric value in 1 gram of each macronutrient.
4. Explain the types, functions, and digestion of proteins.
5. Discuss protein deficiency and excess disorders.
6. Explain the types, functions, and digestion of fats.
7. Discuss fat and lipid deficiency and excess disorders.
8. Explain the types, functions, and digestion of carbohydrates and fiber.
9. Discuss carbohydrate deficiency and excess disorders.
10. Differentiate between fat-soluble and water-soluble vitamins.
11. Identify sources and functions of vitamins, in addition to symptoms of deficiencies and excesses of each.
12. Differentiate between major and trace minerals.
13. Identify sources and functions of minerals, in addition to symptoms of deficiencies and excesses of each.
14. Discuss the functions and importance of water in the body.
15. Discuss how nutritional needs vary over the life span, including the risks and prevention of being overweight or obese.
16. Discuss information found in the Connection features in this chapter.
17. Identify specific safety features.

CRITICAL THINKING CONNECTION

Clinical Assignment

You are caring for a 68-year-old male who has been hospitalized for extreme constipation. He is having bowel movements now, but he has had a poor diet that contributed to his constipation problems. His wife died 9 months ago. He eats mostly canned soups and processed lunch meats on white bread because he does not cook much. He sometimes gets fast food, such as hamburgers or fried chicken. He likes meat but does not eat vegetables or fruits. He tells you they go bad before he can eat them all, so they are a waste of money. He mostly drinks soft drinks throughout the day.

Critical Thinking Questions

1. How will you determine the effectiveness of the current treatment for constipation?
2. What nutritional concerns do you have for this patient?
3. How will you help him manage this issue?
4. How will you help this patient prevent rehospitalization for constipation in the future?

Good nutrition is a primary key to optimal health for each individual in any circumstance. Does it prevent or cure all diseases? Of course not, but it allows the body access to the nutrients needed to supply the energy for all body systems to work together for the good of the body. Energy is the product of *metabolism*, the chemical and physical processes in which body cells break down and use food, water, and other chemicals needed to maintain life and the function of each body system. The breaking-down phase of metabolism is known as **catabolism**, in which the body reduces complex substances down to the simpler substances and sometimes releases energy as a result. When the body uses components to build or reconstruct new components or tissue, it is called **anabolism**.

Proper nutrition is essential for a stronger immune system, less illness, and better health. Even a relatively small nutrient imbalance can cause severe health problems. This chapter will show you the importance of nutrition and specific nutrients pertinent for optimal health. You will learn how nutritional needs differ through the life span, sources of nutrients, and how culture influences individual food preferences.

NUTRITIONAL GUIDELINES

Nutrients are provided through natural (food) and artificial (supplements) sources. Although each nutrient has its own specific functions, they all must work together to provide optimal functioning of the body systems.

Food provides energy, which is measured in kilocalories (kcal), and this energy can either be used instantly or stored to be used at a later time. The number of kilocalories that is required to function optimally varies depending on age, gender, and activity levels ([Table 23.1](#)).

KNOWLEDGE CONNECTION

How many calories does the average active adult male need per day? An active young child? Which would require intake of more calories: active adolescent male or female?

FDA Guidelines and Food Labels

The Food and Drug Administration (FDA) first established standards for the minimum amount of nutrients necessary to protect most individuals from disease. These standard guidelines are known as *dietary reference intakes (DRIs)* and contain recommendations for sex, age, pregnancy, and lactation. The DRIs for individuals and for various nutrients can be viewed on the government Web site at <https://www.nal.usda.gov/fnic/dri-tables-and-application-reports>. Various laws ensure that Nutrition Facts food labels are found on all processed food products. The Nutrition Facts label has a goal of making it easier for consumers to make informed food choices ([Fig. 23.1](#)). Each label provides an ingredient list and nutritional descriptions, including the following:

- The amount of calories in a product are printed in a large, bold, easy-to-read font

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- The amount of calories in a product are printed in a large, bold, easy-to-read font.
- The serving size and servings per container are easy to read, and serving sizes reflect the amount people typically consume at one time.
- Labels show “added sugars” in grams; these are sugars that were added during the processing of the food.
- The labeling breaks out vitamin D, calcium, iron, and potassium in actual amounts and gives the percentages of the daily recommended amounts provided in the product. These micronutrients are included on food labels because Americans are typically deficient in these specific micronutrients.

Dietary Guidelines for Americans 2020-2025

To assist the public to eat nutritionally and reach optimal health, the U.S. Department of Agriculture (USDA) and the Department of Health and Human Services (DHHS) have jointly created and published recommended dietary guidelines. They review these guidelines every 5 years and update them as necessary; the newest edition is the *Dietary Guidelines for Americans 2020-2025*. These guidelines may be viewed at the government Web site at <https://www.dietaryguidelines.gov/>.

The key recommendations of these guidelines are that Americans should consume a healthy eating pattern that includes the following:

- A variety of vegetables, including dark green, red, and orange; legumes (beans and peas); and starchy and other types of vegetables
- Fruits, especially whole fruits
- Grains (at least half of which are whole grains)
- Fat-free or low-fat dairy, such as milk, yogurt, cheese, and fortified soy beverages
- A variety of protein foods, such as seafood, lean meats, poultry, eggs, legumes, nuts, seeds, and soy products
- Oils

• WORD • BUILDING •

metabolism: metabol - change + ism - principle
catabolism: cata - down + bol - throw + ism - principle
anabolism: ana - up + bol - throw + ism - principle

Nutrition Facts	
8 servings per container	
Serving size	2/3 cup (55g)
Amount per serving	
Calories	230
% Daily Value*	
Total Fat 8g	10%
Saturated Fat 1g	5%
Trans Fat 0g	
Cholesterol 0mg	0%
Sodium 160 mg	7%
Total Carbohydrate 37g	13%
Dietary Fiber 4g	14%
Total Sugars 12g	
Includes 10g Added Sugars	20%
Protein 3g	
Vitamin D 2mcg	10%
Calcium 260mg	20%
Iron 8mg	45%
Potassium 235mg	6%

* The % Daily Value (DV) tells you how much nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

FIGURE 23.1 Nutrition facts food labels (from Food and Drug Administration. The New Nutrition Facts Label: What’s In It For You? Available at: <https://www.fda.gov/food/nutrition-education-resources-materials/new-nutrition-facts->

<https://www.fda.gov/1000/nutrition-education-resources-materials/new-nutrition-facts-label>. Accessed March 13, 2021.).

Table 23.1

Estimated Calorie Requirements (in Kilocalories) by Sex and Age Group at Three Levels of Physical Activity

Sex	Age (Years)	Activity Level		
		Sedentary	Moderately Active	Active
Child, Female and Male	2-3	1,000	1,000-1,400	1,000-1,400
	Female			
	4-8	1,200-1,400	1,400-1,600	1,600-1,800
	9-13	1,400-1,600	1,600-2,000	1,800-2,200
	14-18	1,800	2,000	2,400
	19-25	2,000	2,200	2,400
	26-50	1,800	2,000	2,200-2,400
Male	51+	1,600	1,800	2,000-2,200
	4-8	1,200-1,400	1,400-1,600	1,600-2,000
	9-13	1,600-2,000	1,800-2,200	2,000-2,600
	14-18	2,000-2,400	2,400-2,800	2,800-3,200
	19-25	2,400-2,600	2,800	3,000
	26-50	2,200-2,400	2,400-2,600	2,800-3,000
	51+	2,000-2,200	2,200-2,400	2,400-2,800

Source: U.S. Department of Agriculture and U.S. Department of Health and Human Services. *Dietary Guidelines for Americans 2020-2025*, 9th ed. Available at: <https://www.dietaryguidelines.gov/>. Accessed March 10, 2021.

In addition, these guidelines recommend that Americans limit saturated fats, trans fats, added sugars, and sodium in their diets. Less than 10% of calories should come from added sugars, less than 10% should come from saturated fats, and sodium should be less than 2,300 mg per day. Alcohol may be consumed in moderation, which means one drink per day for women and two drinks per day for men. The Dietary Guidelines for Americans also provide healthy eating patterns for a Mediterranean diet and for a vegetarian diet.

Current eating patterns in the United States show that three-fourths of us do not consume enough vegetables, fruits, dairy, and oils. More than three-fourths of us meet or exceed the total grain and total protein recommendations but are not meeting the requirements for subgroups of proteins, such as seafood, legumes, nuts, and soy products. And most Americans exceed the recommendation for added sugars, saturated fats, and sodium. Three-fourths of all adults and nearly one-third of children and youth in the United States are overweight or obese.

Women of Childbearing Age

Safety: Because fetal development of the brain and spinal cord occurs within the first few weeks of pregnancy, one of the USDA and DHHS' key recommendations is that women of childbearing age consume 200 mcg of synthetic folic acid daily from fortified foods and/or supplements in addition to the amounts of food folate contained in a healthy eating pattern. Food sources include all enriched grains, beans, peas, oranges, orange juice, and dark green leafy vegetables. These recommendations are intended to prevent neural tube defects, such as spina bifida, in the developing fetus. [Box 23.1](#) provides nutrition recommendations for pregnant women.

KNOWLEDGE CONNECTION

Why are vitamin D, iron, calcium, and potassium included on food labels? What are "added sugars"? How much added sugar should be included in a healthy eating pattern? List sources of protein in addition to beef and chicken. What dietary recommendations are made for women who are of childbearing age or who are pregnant?

Box 23.1

Box 23.1

Nutrition Recommendations During Pregnancy

The American College of Obstetricians and Gynecologists recommends that pregnant women double the amount of iron that they consumed pre-pregnancy. This is recommended because more iron is required to carry oxygen to the fetal tissues. Folic acid also must be increased to prevent birth defects of the nervous system. Most physicians recommend that pregnant women take prenatal vitamins to ensure the adequate intake of vital nutrients and to help provide these extra nutrients. The increased needs for iron and folic acid remain during lactation as well. During pregnancy, the only artificial sweeteners that the FDA has approved are aspartame, sucralose, and acesulfame-K. Other recommendations for pregnancy include an increase of 340 calories per day and 450 calories daily for breastfeeding mothers.

The USDA [ChooseMyPlate.gov](https://www.choosemyplate.gov) has health and nutrition information for pregnant and breastfeeding mothers. They include a Daily Food Plan for Moms that calculates the increased nutritional needs for the mother and the mother-to-be based on data entered by the individual.

MyPlate

The USDA's Center for Nutrition Policy and Promotion uses a visual symbol called MyPlate, along with numerous other materials, to help guide Americans to healthier eating ([Fig. 23.2](#)). In essence, half of your plate of food should be fruits and vegetables, with veggies making up the greater portion of the two groups. Grains and protein should make up the other half of the plate, with the grains portion being the larger of the two groups. The dairy serving should be small and consist of low-fat or nonfat milk or small servings of other low-fat dairy products. Depending on the age of the individual, the entire daily intake of dairy should be restricted to a total of 1 ½ to 3 cups of low-fat or nonfat milk, cheese, yogurt, or soymilk. (Refer to [Box 23.2](#) for additional tips for healthy nutrition. To view all information associated with MyPlate, go to www.choosemyplate.gov.)

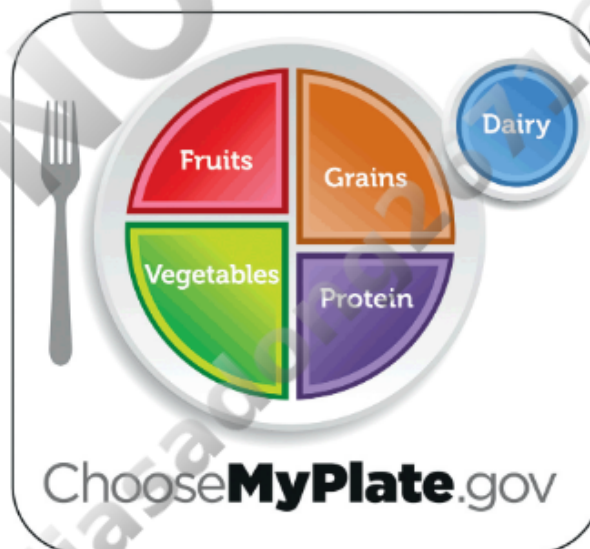


FIGURE 23.2 The USDA's MyPlate food management system (from U.S. Department of Agriculture. ChooseMyPlate. Available at: www.choosemyplate.gov. Accessed March 10, 2021.).

Box 23.2

ChooseMyPlate Nutrition Tips

- Make half of your plate fruits and vegetables
- Focus on whole fruits
- Vary your vegetables
- Make half of your grains whole grains
- Move to low fat or fat free milk or yogurt

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- Move to low-fat or fat-free milk or yogurt
- Vary the kinds of proteins you consume

Source: www.ChooseMyPlate.gov.

Anatomy and Physiology Connection

Digestion

Digestion is the process by which food is broken down in the gastrointestinal (GI) tract, releasing nutrients for the body to use. The process of digestion begins in the mouth, where a digestive enzyme called *salivary amylase* begins breaking down simple carbohydrates and fat. The mechanical motion of chewing breaks down starches into smaller products, with saliva moistening the foods. The esophagus transports foods from the mouth into the stomach but does not participate in chemical or mechanical digestion. Muscles and nerves work together to perform *peristalsis*, moving the food from the esophagus to the stomach, into the small intestine, into the large intestine, and finally to the rectum.

Once the food enters the stomach, the gastric juices of the stomach, which are acidic in nature, *hydrolyze* (adds water to break down) carbohydrates into maltose and sucrose. Proteolytic and lipolytic enzymes mix with food and break down small amounts of proteins and fat. The stomach gently contracts and expands to further break down food particles. Serving mostly as a reservoir for the food, little absorption occurs here except for alcohol. As peristalsis gently propels the food bolus from the stomach into the small intestine, the digestive processes in the small intestine are activated.

Most digestion occurs in the first several feet of the small intestine. Digestive enzymes are released from the liver, gallbladder, pancreas, and small intestine, resulting in the breakdown of starches and proteins. Dietary fats are reduced to free fatty acids and monoglycerides. Almost all macronutrients, minerals, and vitamins, along with some fluid, are absorbed at this point in the small intestine. **Absorption** is the process by which nutrients are taken from the end products of digestion into the villi (small projections of intestinal mucosa that line the walls of the small intestine) that contain capillaries. This is the point where the majority of nutrients are absorbed into the bloodstream.

The components of food, mostly cellulose, that are not absorbed in the small intestine are now moved by peristalsis into the large intestine. The colon and rectum absorb most of the fluid produced by the digestive process and some electrolytes into the bloodstream. The remaining bulk is expelled from the rectum as fecal material.

The nutrients absorbed from the GI tract travel through the portal vein to the liver, where they are released into circulation, transformed into other products, or stored. Once released into circulation, nutrients are delivered via the bloodstream or lymphatic system to areas of need. Certain nutrients, such as fatty acids, that are not immediately used are stored in the body as adipose tissue.

ESSENTIAL NUTRIENTS

Nutrients are divided into two major categories: macronutrients and micronutrients. Macronutrients are required in larger amounts on a daily basis and include the following:

- Protein
- Carbohydrates
- Fats
- Water

Micronutrients include the following:

- Vitamins
- Minerals (which are needed in smaller amounts, known as *trace minerals*)

For optimal health, it is important to ingest all the essential nutrients, including water, in a balanced ratio. **Safety: Although most people understand that an inadequate intake of nutrients can lead to health problems and less-than-optimal body functioning, it is important to realize that too much of certain nutrients can be just as problematic.** It will be your job to understand the problems of both nutrient deficiencies and excesses so that you will be prepared to educate your patients.

Protein

Protein is the primary building block of the body to make new cells and to help wounds heal. Most of the organs and muscular and integumentary systems are constructed of protein. A gram of protein

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The organs and muscular and integumentary systems are constructed of protein. A gram of protein contains 4 kcal, and in a well-balanced diet, protein should provide between 10% and 35% of the total daily calories. Most adults require a daily intake of 50 to 60 g of protein.

Protein Functions

The functions of protein include the following:

- New tissue production
- Formation of antibodies, enzymes, and hormones
- Necessary component for heat and energy production
- Component to assist in maintaining fluid balance between cells and bloodstream
- Maintaining acid-base balance

Protein Types

The human body contains more than 100,000 types of protein. Approximately 50% of this protein is found in skeletal muscle tissue. These proteins are composed of three types of amino acids: (1) nonessential, (2) conditional, and (3) essential (there are nine of these, and they contain the nitrogen needed by the body to build new tissue). The body continuously breaks down these proteins into its smaller components (amino acids) using the process of catabolism. Then through anabolism, these amino acids are reorganized, or built, into tissue products.

Nonessential amino acids are produced by the liver and thus are not essential to be included in our diet. Conditional amino acids are those that become essential *only* when an individual is stressed or ill with certain conditions. Dietary-wise, **essential amino acids** are the most important: They must be obtained from food sources because the body is unable to produce them. The proteins that contain all nine of these essential amino acids (along with some of the nonessential amino acids) are known as **complete proteins**. They come from animal and plant sources such as eggs, cheese, milk, sesame, peanuts, whitefish, pork, and beef. Soy protein is the only source of complete proteins that is not an animal source.

• WORD • BUILDING •

hydrolyze: hydro - water + lyze - dissolve

proteolytic: proteo - protein + lytic - able to dissolve

lipolytic: lipo - fat + lytic - able to dissolve

Proteins that do not contain all nine of the essential amino acids are termed **incomplete proteins** and come from plant sources. They include corn, wheat, seeds, beans, brown rice, and nuts. It is important to note, though, that eating *combinations* of incomplete proteins provides the body with all nine amino acids needed for complete protein synthesis. By consuming two or more of the incomplete protein sources, an individual can still meet his or her protein requirements. Commonly used combinations include the following:

- Red beans and brown rice
- Peanut butter and whole wheat bread
- Brown beans and cornbread

Protein Digestion

Although much of protein digestion occurs in the small intestine, it begins in the stomach, where hydrochloric acid breaks down large proteins into amino acids, which are partially digested by the enzyme pepsin. Once the food moves from the stomach into the duodenum, trypsin works to further digest the amino acids so that they can be absorbed into the small projections of the intestinal wall, known as *villi*. Networks of capillaries are found inside the villi, providing nutrients access to the bloodstream. Now the blood distributes the amino acids to the entire body so that they can be used for anabolism, with the goal of maintaining a balance of protein between the blood and tissues.

Protein Deficiency and Excess Disorders

Because of poverty, a lack of education, and a lack of farming resources, protein deficiencies are more commonly found in resource-limited countries than in the United States. When they are found in the United States, they are generally the result of poverty, but they may be seen in the higher

in the United States, they are generally the result of poverty, but they may be seen in the higher socioeconomic populations because of a simple failure to eat adequate protein. Severe protein deficiency results in a disease known as *kwashiorkor*. It causes severe emaciation, a grossly edematous (swollen) abdomen caused by ascites (fluid collection in the peritoneal cavity) and liver enlargement, lethargy, failure to grow, and skin infections. If not treated early, limitations in mental development will also occur.

The consumption of excess protein does not result in additional muscle building. The excess protein is metabolized and stored as fat to be used at a later time when an additional energy source is required. Excess protein can play a role in the worsening of kidney disease. Protein intake and the removal of nitrogen (a waste product of protein digestion) by the kidneys should be in balance in most situations. However, there are times when retained nitrogen needs to be more or less than at other times. (See [Box 23.3](#) for information on nitrogen balance.)

Box 23.3

Nitrogen Balance

The amount of nitrogen that is ingested in the form of protein should balance with the level of nitrogen that is used by the body and excesses that are removed from the body by the kidneys. When these levels are equal, it is called **nitrogen balance**. When the intake of nitrogen is larger than the amounts excreted in the urine, there is nitrogen left over to build new tissue. This is known as a state of positive nitrogen balance and is necessary during the growth of a fetus during pregnancy; the quick growth spurts during childhood; and during times of repair, as in healing large burns or wounds.

Negative nitrogen balance occurs when excess nitrogen is excreted, with inadequate amounts of protein being consumed in the diet. Negative nitrogen balance occurs commonly in malnutrition. This condition can cause a loss of muscle tone, weight loss, failure to heal or repair tissue, and growth retardation. Nitrogen balance is measured in blood urea nitrogen in the blood and urea urine nitrogen levels in a 24-hour urine test.

KNOWLEDGE CONNECTION

How many calories are there in a gram of protein? How many grams of protein should the average adult ingest daily? What are the functions of protein? Explain the difference between complete and incomplete proteins. Which type of protein is found in animal sources? Plant sources? Give two examples of food combinations that will provide complete proteins.

Fats/Lipids

While fat may seem like something that we want to avoid, it is actually a necessary macronutrient needed for a variety of functions in the body. Unlike the other two macronutrients, fats provide 9 kcals per gram ingested, so they are twice as high in calories per gram as protein and carbohydrates. Knowing which fats are most beneficial and why will help you choose fat calories carefully.

Types of Fats and Lipids

Some fats and lipids are manufactured and stored by the body, and some are obtained through foods. Our livers manufacture a fat or lipid called *cholesterol*. It is a necessary component for good health, but it must be kept within certain limits or it can become a health hazard that leads to various disease processes.

• WORD • BUILDING •

lipoprotein: lipo – fat + protein

Cholesterol is transported throughout the body by low-density lipoprotein (LDL) and high-density lipoprotein (HDL). The LDL is commonly known as *bad cholesterol* because it transports the cholesterol to the cells from the liver, and too much is a bad thing. If the cells do not use all of the

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cholesterol to the cells from the liver, and too much is a bad thing. If the cells do not use all of the cholesterol that is carried to them, the cholesterol can build up and lodge in artery walls. HDL is the transporter responsible for taking the excess cholesterol back to the liver from the cells, helping to maintain a proper balance. Thus its common name is *good cholesterol*.

Triglycerides are a form of fat that is stored in the body but is also found in food. They are made from any combination of three fatty acids. Any time excess fats are ingested, they may be stored as triglyceride within the fat cells. An excessively high level of triglycerides contributes to coronary artery disease and other metabolic syndromes.

Laboratory values for total cholesterol, LDL, HDL, and triglycerides are grouped into a *lipid profile* (Table 23.2). A trick for remembering which cholesterol is desired at higher ranges and which one at lower ranges is that the LDL (bad cholesterol) should be *low* and the HDL (good cholesterol) should be *high*.

DIETARY FATS. When one hears the term *dietary fats* or *lipids*, it is commonly in connection with a negative perspective, but a healthy diet must include a certain amount of fat.

Dietary fat is broken down into several categories, including saturated, monounsaturated, polyunsaturated, and trans fats. Some fats are good for us, while others are implicated with cardiovascular disease because they raise cholesterol levels and are associated with type 2 diabetes. This makes it important that you learn the differences between them. The healthiest fats to include in the diet are the monounsaturated and the polyunsaturated fats.

MONOUNSATURATED AND POLYUNSATURATED FATS. Monounsaturated fats become solid when they are refrigerated but are liquid when they are kept at room temperature; they include olive oil, canola oil, and peanut oil. They are better for us because they help to lower our blood levels of the bad cholesterol, LDL.

Polyunsaturated fats are found in sources such as corn oil, soybean oil, sesame oil, and safflower oil. Two examples of essential polyunsaturated fats are the omega-3 and omega-6 fatty acids that are found in several types of fatty fish, walnuts, and flaxseed. These are known to decrease one's risk of type 2 diabetes and cardiovascular disease.

SATURATED FATS. *Saturated fats* come from animal sources, are solid at room temperature, and are implicated in cardiovascular disease because they raise blood cholesterol. Thus these fats should be limited in a healthy diet. Although most saturated fats are found in animal sources such as meat, whole milk, and cheese, some are present in coconut and palm oils, hydrogenated vegetable oil, and cocoa butter, which are common ingredients in packaged foods such as cookies, cakes, and pies.

TRANS FATS. *Trans fats* are vegetable oils that have been partially hydrogenated, a process of adding hydrogen, which changes the unsaturated fat to saturated fat. Trans fats are used to keep foods fresh and are found mostly in processed and fast foods. They have been linked to increasing blood cholesterol levels and are being slowly removed from processed foods as directed by the FDA.

Table 23.2

Lipid Panel Laboratory Value Ranges for Adults

Test Component	Desired Level	Near Desired Level	Borderline	Health Risk
Total cholesterol	Less than 200 mg/dL	N/A	200–239 mg/dL	More than 240 mg/dL high
Low-density lipoprotein (LDL) (bad)	Less than 100 mg/dL	100–129 mg/dL	130–159 mg/dL	More than 160–189 mg/dL high, >190 mg/dL very high
High-density lipoprotein (HDL) (good)	More than 60 mg/dL	40–60 mg/dL	N/A	Less than 40 mg/dL low
Triglycerides	Less than 150 mg/dL	N/A	150–199 mg/dL	More than 200–499 mg/dL high, >500 mg/dL very high

Functions of Fats

Cholesterol, which is manufactured by the body, is required to:

- Form each of the body cell membranes of every system
- Produce the adrenal and sex hormones in addition to bile
- Protect our nerves
- Convert sunshine to vitamin D
- Help metabolize fat-soluble vitamins

Dietary fats that are ingested are needed by the body; they are an essential concentrated source of fatty acids and energy and are necessary for healthy skin. In addition, dietary fats are required to help the body absorb fat-soluble vitamins. They are stored as fatty (adipose) tissue, helping to support the internal organs and insulating the body from environmental temperature extremes.

Fat Digestion

Lingual lipase starts working on the fat in the mouth. Then food is swallowed and remains in the stomach longer than carbohydrates, which provides a longer sense of feeling full. Gastric lipase and gastric churning help to break down the fats before they move into the small intestine, where the bulk of fat digestion occurs. There the fats are emulsified, or broken down into tiny droplets, by bile acids from the gallbladder. Pancreatic lipase changes the fat molecules into fatty acids and monoglycerols, which are absorbed into the bloodstream through the intestinal villi for use throughout the body. Fatty acids and glycerol are used in the production of phospholipids, the primary component of cell membranes for all body systems. Any fats that are not used are stored as adipose (fatty) tissue.

Fat/Lipid Deficiency and Excess Disorders

Too little fat intake is rare in the United States. However, if a person is eating a severely fat-restricted diet for a long period, it is possible for that individual to become deficient in fat-soluble vitamins because they are stored in the body fat.

The American Heart Association recommends that no more than 25% to 35% of one's total daily intake of calories should come from fat to help prevent coronary artery disease and other metabolic syndromes. It is important to note that the majority of one's consumed fat should be the healthier fats rather than saturated fat. An overindulgence of saturated fats or eating more than 35% of the daily caloric intake in the form of fat is known to contribute to being overweight and obese. Remember that fat provides twice as many calories per gram ingested as protein and carbohydrates.

KNOWLEDGE CONNECTION

How many calories does a gram of fat provide? Which type of cholesterol is desired in higher ranges? Which of the types of fat are considered to contribute to cardiovascular disease? List several sources of trans fat, saturated fat, monounsaturated fat, and polyunsaturated fat. Which type of fat is known to decrease the bad cholesterol known as LDL? What are some food sources that a person might consume to help keep LDLs in the lower range?

Carbohydrates

Carbohydrates provide 4 kcal of energy per gram. As with fat, overindulgence in carbohydrates contributes to excessive weight gain and obesity.

Types of Carbohydrates

Carbohydrates are classified as either simple or complex. **Simple carbohydrates**, also known as *simple sugars*, are chemically made up of one or two sugar molecules that are absorbed rapidly. The following are four examples of simple sugars:

- **Glucose:** our primary energy source, the most common form
- **Fructose:** fruit sugar, the sweetest natural sugar
- **Sucrose:** table sugar

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- **Lactose:** milk sugar, the least-sweet natural sugar

Sources of simple carbohydrates include fruit juice, honey, milk, white bread, table sugar, syrup, candy, pastries, and colas. The starches and fiber (cellulose) make up **complex carbohydrates**. Chemically, they differ from the simple carbohydrates because they have long chains of three or more molecules of sugar, so they take a little longer to digest, absorb, and produce energy for the body. Food sources of complex carbohydrates can be found in [Box 23.4](#).

Box 23.4

Food Sources of Complex Carbohydrates

Potatoes, white and sweet	Oatmeal and rolled oats
Brown rice	Whole grain wheat bread
Millet, buckwheat, and bulgur	Pasta
Broccoli	Whole fresh fruit
Beans and legumes	Peas and lentils
Whole grain, high-fiber, low-sugar cereals	

One of the main differences between simple and complex carbohydrates is their effects on the blood glucose level. Simple carbohydrates, such as glucose and fructose, cause a rapid rise in the blood glucose level. The high blood glucose level then stimulates the pancreas to produce more insulin, which will rapidly decrease blood glucose and increase appetite, and thus increase eating. When this occurs excessively, we put on extra pounds. It is wiser to consume more complex carbohydrates than simple carbohydrates because they help us to feel full longer and provide a more stable blood glucose level.

Functions of Carbohydrates

Carbohydrates serve as the primary source of energy for the body, particularly the cardiovascular, musculoskeletal, and nervous systems. They are more quickly digested into a useable form for energy (glucose) than either proteins or fats.

The USDA recommends that 45% to 65% of one's total calorie consumption come from complex carbohydrates, but there is no actual recommendation for how much of the diet should come from simple carbohydrates, just that they should be limited. Excess glucose from carbohydrate breakdown normally is stored in the liver and skeletal muscles in the form of glycogen, to be used at a later time as an energy source. Without adequate consumption of complex carbohydrates, the readily available glucose and the stores of glycogen will be insufficient for energy production. Thus the body will resort to burning fat and protein for energy, reducing the available protein for repairing damaged tissue and growing new tissue.

• WORD • BUILDING •

phospholipids: phos - light + pho - carry + lipids - fat

Carbohydrate Digestion

As we just mentioned, the starchy carbohydrates are reduced to glucose during digestion. Salivary amylase in the mouth, along with amylase and maltase in the small intestine, break down the starches to glucose. Very little digestion of starch occurs while it is in the stomach. In the duodenum, absorption takes place through the intestinal wall villi into the bloodstream, where it is transported directly to the cells for immediate use, or to the liver for storage as glycogen.

Carbohydrate Deficiency and Excess Disorders

No known carbohydrate deficiency disease exists. However, when a person does not eat enough glucose for the body's energy needs, the body burns stored fat. When this goes on for extended periods in a patient with diabetes, it can cause a serious elevation in blood acid levels, called diabetic ketoacidosis (DKA). This is seen in type 1 diabetes when the body cannot access glucose for energy.

for energy.

Excessive carbohydrate ingestion commonly leads to weight gain and obesity. Eating excessive carbohydrates raises the blood glucose; this in turn raises the insulin level, which then lowers blood glucose. A drop in one’s glucose level results in hunger and food cravings, often for simple carbohydrates in the form of junk food. The more food that is consumed, especially simple carbohydrates, the more calories that must be burned to prevent weight gain.

Fiber

Fiber is included in our discussion of carbohydrates because it is a type of carbohydrate that the body cannot digest.

TYPES OF FIBER. Dietary fiber comes in two forms: soluble and insoluble. As soluble fiber is digested, it attracts water to form a gel-like substance, which remains in the stomach longer, providing the individual with a sense of feeling full for a longer period. The insoluble fiber is not digested but passes through the intestines, acting like a bulk laxative and drawing water into the intestines, helping to prevent constipation.

FUNCTIONS OF FIBER. A diet that includes adequate levels of fiber has numerous additional health benefits beyond the stabilization of blood glucose levels:

- A decrease in LDL cholesterol, which reduces the risk of coronary artery disease
- Promotion of normal bowel function and the prevention of constipation
- Increased absorption of minerals
- A lowered colon pH, which helps to discourage pathogen and cancer cell growth
- Support of GI tract normal flora by providing them with a food source
- Promotion of weight loss

For optimal health, the average adult needs 25 to 35 g of fiber every day, but few people in the United States consume more than half of this amount. This contributes to numerous health issues that are so easily prevented. Teach your patients about the health benefits of fiber, the food sources of fiber, and the amount to ingest daily. However, it is very important to explain that when increasing fiber intake, a person must do so gradually or they will suffer *extreme problems with flatus*. Explain that fiber should be increased slowly over 2 to 4 weeks to give the GI tract time to adjust to the sudden increase. Refer to [Table 23.3](#) for food sources of soluble and insoluble fiber.

KNOWLEDGE CONNECTION

How many calories per gram are in carbohydrates compared with fats and protein? Explain the difference between simple carbohydrates and complex carbohydrates. What percentage of total daily caloric intake should come from carbohydrates? List at least four food sources of fiber. What is the recommended daily intake of fiber?

Vitamins

Safety: It is vital to consume an adequate amount of most vitamins in our daily diet (or through supplements) because the body cannot manufacture many of them. With exposure of the skin to sunlight 15 minutes twice per week, the body can synthesize a significant amount of the needed vitamin D. The normal flora of the intestines also helps by producing vitamin K, but most of our vitamins are obtained from plant sources and animal products. They are a crucial component for the body’s metabolic processes, such as synthesis of hemoglobin, conduction of nerve impulses, bone and collagen development, and blood clotting. There are two categories of vitamins: water soluble and fat soluble.

Table 23.3

Food Sources of Fiber	
Soluble	Insoluble
• Oat bran • Beans • Rice bran	• Brown rice • Whole wheat • Bran

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• Rice bran	• Bran
• Oatmeal	• Dark green leafy vegetables
• Apples	• Nuts
• Strawberries	• Seeds
• Oranges	• Cucumbers
• Pears	• Cabbage
• Blueberries	• Tomatoes
• Dried peas	• Broccoli
• Celery	• Grapes/raisins
• Carrots	• Fruit

Water-soluble vitamins are dissolvable in water and therefore can be absorbed directly into the bloodstream from the GI tract. They cannot be stored in the body, and some of them are lost in the urine, thus the necessity of consuming them on a daily basis. Water-soluble vitamins include the following:

- B vitamins
 - B₁ (thiamine)
 - B₂ (riboflavin)
 - B₃ (niacin)
 - B₄ (adenine)
 - B₅ (pantothenic acid)
 - B₆ (pyridoxine)
 - B₇ (biotin)
 - B₈ (inositol)
 - B₉ (folic acid)
 - B₁₀ (para-aminobenzoic acid)
 - B₁₁ (choline)
 - B₁₂ (cyanocobalamin)
- Vitamin C (ascorbic acid)

Among other functions, the B vitamins serve as coenzymes for metabolism throughout the body, and they help to stimulate appetite, maintain healthy nervous and integumentary systems, and assist with growth. Vitamin C is essential in immune system functioning, helps the body absorb iron, and is vital for wound healing. Some vitamins are used to treat specific disease conditions. Vitamin B₃ (niacin) is used to treat hyperlipidemia, and vitamin B₆ (pyridoxine) is used to prevent peripheral neuropathy in patients taking the drug isoniazid (INH) for tuberculosis.

Fat-soluble vitamins A, D, E, and K are dissolvable only in fat, and, unlike water-soluble vitamins, they can be stored in body fat when there is more ingested than the body needs at that specific time. **Safety: Because fat-soluble vitamins can be stored, they can become toxic when consumed in excessively large doses over a long period.**

Vitamin Deficiencies and Excesses

Vitamin deficiencies in the United States are generally the result of malabsorption disorders that prevent the vitamins from being absorbed and used by the body. Examples of such disorders include Crohn's disease, celiac disease, and pancreatitis. Vitamin deficiencies may also be the result of poor nutritional habits.

Safety: Vitamin excesses are usually caused by overuse or inappropriate use of dietary supplements. Refer to [Table 23.4](#) for information about deficiencies and excesses of vitamins.

Antioxidants

Physical and emotional stress and environmental factors such as poor diet, pollution, certain

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Physical and emotional stress and environmental factors such as poor diet, pollution, certain medications, and alcohol can cause body cells to react with molecules of oxygen in a way that produces unstable molecules called *free radicals*. These free radicals can cause damage to DNA and protein, which can lead to various diseases such as cancer. Vitamins A, C, and E are known as **antioxidants** because they have been identified as helping to deactivate these free radicals before they can cause damage. Ultimately, the antioxidants help to promote health and prevent disease. [Table 23.4](#) provides information about vitamin functions, sources, deficiencies, and excesses.

KNOWLEDGE CONNECTION

Which vitamins are water soluble? Which are fat soluble? Which vitamins are toxic when consumed in excessive doses? Explain the function of antioxidants. Relate at least three functions of the B vitamins. Which vitamin is important for growing new tissue, such as is required after surgery? Which vitamins are antioxidants?

Minerals

Minerals are inorganic compounds that come from the earth, and because they cannot be digested, they retain their original structure once ingested. They are classified as either major or trace minerals. Major minerals are those that we must ingest a minimum of 100 mg/d for optimal health, and trace minerals are those that we require less than 100 mg/d. The most abundant mineral in the body is calcium. Up to 99% of body calcium is stored in the bones, leaving only 1% to circulate in the bloodstream to be used by the rest of the body. Minerals are used by all body tissues for numerous functions, including the following:

- Forming the structure of the hard parts of the body (bones, teeth, nails)
- Assisting in water metabolism and fluid and electrolyte balance
- Activating enzymes and hormones
- Assisting in acid-base balance
- Nerve cell transmission
- Muscle contraction

For a breakdown of major and trace minerals, refer to [Box 23.5](#). Major functions and sources of minerals are listed in [Table 23.5](#).

Mineral Deficiencies and Excesses

Mineral deficiencies do not normally develop in people who routinely eat well-balanced diets. However, following a restrictive diet can result in a shortage of one or more minerals. Someone who does not use iodized salt and does not like any type of seafood will be at risk for developing an iodine deficiency. **Safety: Vegetarians may be at risk for iron deficiency unless they are careful to ingest a higher level of iron-rich, nonmeat foods such as beans, dried fruit, whole grains, and molasses.**

Mineral excesses may be the result of improper use of supplements in adults. The accidental ingestion of iron supplements by infants and children can lead to serious iron toxicities. A genetic abnormality known as *hemochromatosis* is a disease in which the person absorbs and accumulates excessive stores of iron. Another genetic abnormality, Wilson disease, causes the body to absorb an excess of copper from the intestine, which then accumulates in the corneas, brain, liver, and kidneys. Refer to [Table 23.5](#) for information on the effects of mineral deficiencies and excesses.

• WORD • BUILDING •

hyperlipidemia: hyper - excessive + lipid - fat + emia - blood

Table 23.4**Vitamin Deficiencies and Excesses**

Vitamin	Source	Physiological Functions	Deficiency	Excess
Fat-Soluble Vitamins A (two forms): retinol and beta carotene	Liver, pumpkin, sweet potatoes, whole milk, cheese, carrots, egg yolk, cod, and cantaloupe	Beta carotene is an antioxidant. Helps to form skin, bone, teeth, and mucous membranes and keeps them healthy. Promotes night vision.	Infections and vision problems	Nausea, dizziness, headache, dermatitis, coma, and death
D	Vitamin D-fortified dairy products, fortified margarine, fatty fish, beef liver, and eggs. (The body can make vitamin D using the energy of sunlight on the skin.)	Helps to harden bones and teeth. Necessary to absorb calcium.	Rickets in children and bone softening and deformity in adults	Rare but serious, nausea, vomiting, anorexia, kidney stones and damage, and mental and physical growth retardation
E	Vegetable oils, fortified cereals, tomatoes, peanut butter, leafy vegetables, avocados, whole grains, sunflower seeds, nuts, and asparagus	Acts as an antioxidant, helping to slow down cell-damaging processes. Helps cells to communicate with each other. Helps protect skin from ultraviolet light damage.	Peripheral neuropathy and ataxia	Normally no side effects, but researchers differ in opinion; some feel it can act as an anticoagulant
K	Made by intestinal bacteria Food sources include spinach, meat, asparagus, beans, eggs, and strawberries	Helps blood to clot.	Excessive bleeding	No toxicity symptoms reported
Water-Soluble Vitamins B ₁ (thiamine)	Beef, liver, pork, whole grains, enriched breads and cereals, powdered milk, nuts, seeds, and dried peas and beans	Provides energy via carbohydrate metabolism. Promotes muscle function and nerve conduction.	Weakness, mental confusion, fatigue, heart problems, paralysis, edema, death, and beriberi and Wernicke disease	No toxicity symptoms reported, excess eliminated in urine
B ₂ (riboflavin)	Eggs, milk and dairy products, broccoli, Brussels sprouts, spinach, mushrooms, and enriched breads and cereals	Antioxidant. Red blood cell production. Metabolism of carbohydrates into glucose.	Fatigue, GI problems, cracks at mouth corners, and swollen magenta-colored tongue <i>Uncommon:</i> dermatitis, light sensitivity, and anemia	No toxicity symptoms reported
B ₃ (niacin)	Meat, beets, yeast, fish, milk, sunflower seeds, and peanuts	Improves circulation. Helps break down fats and protein. Promotes healthy hair, skin, eyes, and liver. Assists adrenal gland to make hormones.	Canker sores, indigestion, vomiting, and fatigue <i>Pellagra: Four Ds</i> • Dermatitis • Diarrhea • Dementia • Death	Flushing caused by capillary dilation, liver damage, gastric ulcers, and nausea and vomiting

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		to make hormones. Promotes healthy nervous system function.	• Death	
B ₆	Meat, fish, poultry, fruits (not citrus), potatoes, liver, and fortified cereals	Assists enzyme functions. Essential in protein metabolism. Promotes healthy nervous system and immune system function.	Can be asymptomatic Dermatitis, cracks at mouth corners, glossitis, weakened immune system, depression, and confusion	Peripheral neuropathy (numbness, pain, tingling of extremities)
B ₁₂	Sardines, salmon, cod, scallops, fortified breakfast cereals, yogurt, milk, and veal	Helps develop and maintain nervous system. Essential for red blood cell formation and DNA production. Assists with protein, carbohydrate, and fat metabolism.	Tongue soreness and erythema, dandruff, heart palpitations, pallor, nervousness, peripheral neuropathy, forgetfulness, and macrocytic anemia	No toxicity symptoms reported
C	Citrus fruits, papaya, strawberries, green and red peppers, parsley, turnip greens, Brussels sprouts, cauliflower, and broccoli	Antioxidant. Growth of collagen tissue, skin; especially wound healing. Helps to keep teeth and gums healthy.	<i>Scurvy</i> : bleeding gums, dry skin, petechia, poor wound healing, dry hair, and anemia	Rare but can cause diarrhea, abdominal cramping, and nausea

Box 23.5

Minerals

Six Major Minerals

Calcium
Chloride
Magnesium
Potassium
Phosphorus
Sodium

Nine Trace Minerals

Chromium
Copper
Fluoride
Iodine
Iron
Manganese
Molybdenum
Selenium
Zinc

Note: Both major and trace minerals are equally important for optimal body functioning.

Table 23.5

Functions and Sources of Minerals

Mineral	Function	Sources	Effects of Deficiency and Excess
Calcium	99% found in structure of bones and teeth, 1% in blood used for electrical conduction of nerve impulses in heart, and muscle contraction	Best sources are *dairy products (milk, cheese, cottage cheese, and yogurt), sardines, salmon, spinach, okra, broccoli, beans, almonds, oatmeal, and calcium-fortified juice	<i>Deficiency</i> : Tetany of muscles, bone and tooth loss, osteoporosis <i>Excess</i> : Constipation, flatus, and kidney stones
Chloride	Normal water distribution, electrical balance in cells,	Beef, pork, cheese, *salt, lettuce, celery, and olives	<i>Deficiency</i> : Nausea, vomiting, and diarrhea

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	nerve impulse transmission		
Chromium	Boosts metabolism, helps break down fat and glucose, promotes insulin balance	Green peppers, spinach, liver, beef, eggs, whole grains, bananas, and apples	<i>Deficiency:</i> May elevate blood glucose <i>Excess:</i> Liver damage
Copper	Assists with formation of red blood cells and bones, helps protect cells from damage, helps iron metabolism	Whole grains, mushrooms, beans, nuts, and oysters	<i>Deficiency:</i> Poor growth, fatigue, pallor, and anorexia
Fluoride	Promotes healthy bones and teeth	*Fluoridated drinking water, tea, coffee, and fish from salt water	<i>Deficiency:</i> Tooth decay <i>Excess:</i> Teeth staining and nausea
Iodine	Promotes thyroid function	Saltwater seafood and *iodized salt	<i>Deficiency:</i> Goiter, hypothyroidism, inhibited infant growth, mental sluggishness, and weight gain <i>Excess:</i> Nausea, burning of oral mucosa, diarrhea, inflammation of thyroid, and thyroid cancer
Iron	Essential component of protein/hemoglobin that transports oxygen in the blood; serves as a component of enzymes	Red meats, fish, poultry, whole grains, molasses, dried fruits, lentils, and beans	<i>Deficiency:</i> Pallor, fatigue, shortness of breath, headache, increased irritability, anemia, and fainting <i>Excess:</i> Missed periods, discoloration of skin, and joint swelling and pain
Magnesium	Supports muscle contraction and nerve conduction, electrical conduction of heart, and energy production	Spinach, seafood, bran cereal, legumes, dairy products, and nuts and seeds	<i>Deficiency:</i> May cause elevated blood pressure (BP), agitation, restless leg syndrome, hypotension, muscle spasms, insomnia, and arrhythmias <i>Excess:</i> Increased calcium excretion, weakness, malaise, and GI symptoms
Manganese	Promotes bone formation and metabolism of carbohydrates, cholesterol, and amino acids	Whole grain products, beans, nuts, and tea	<i>Deficiency:</i> Fainting, hearing loss, dermatitis, ataxia, and slowed hair growth <i>Excess:</i> Shaking and tremors, stiff muscles, hypertension, impaired memory, and psychosis
Molybdenum	Assists in metabolism of amino acids	Nuts, grains and grain products, beans, and peas	<i>Deficiency:</i> Difficulty seeing at night, gum problems, tachypnea, and tachycardia <i>Excess:</i> Diarrhea, joint swelling, anemia
Phosphorus	Serves as a structural component of bone and cell membranes	Dairy products, eggs, poultry, red meat, and fish	<i>Deficiency:</i> Anorexia, fatigue, fragile bones, bone pain, bone loss, and increased irritability <i>Excess:</i> Muscle tetany, seizures, and hypocalcemia
Potassium	Needed for proper nerve conduction, especially the heart and muscles, helps to maintain healthy BP level	Bananas, cantaloupe, tomato sauce and puree, white and sweet potatoes, tuna, cod, skim milk, peas, and beans	<i>Deficiency:</i> Muscle weakness and cramping, malaise, fatigue, nausea, vomiting, cramping, numbness, tingling, and life-threatening arrhythmias and cardiac arrest <i>Excess:</i> Anorexia, fatigue, extreme muscle weakness,

Selenium	Antioxidant (reduces oxidation of certain molecules), promotes thyroid hormone function	Seafood and organ meats	extreme muscle weakness, and life-threatening arrhythmias <i>Deficiency:</i> Keshan disease <i>Excess:</i> Hair loss, nail problems, and increased irritability
Sodium	Regulates body compartment fluid levels, acid-base balance; needed for proper nerve conduction and muscle contraction	Salt, celery, beets, dairy products, pork, beef, and <u>*processed</u> foods (Americans' primary source)	<i>Deficiency:</i> Stomach cramps, nausea, vomiting, diarrhea, lethargy, headache, confusion, and seizures caused by brain swelling <i>Excess:</i> Edema and fluid retention, thirst, dry sticky mucous membranes, and elevated BP
Zinc	Necessary for proper growth, supports immune system, serves as a component of over 50 enzymes, essential in wound healing, helps maintain senses of smell and taste	Pork, red meat, lamb, poultry, whole grains, yeast, fortified breakfast cereal, beans, and nuts	<i>Deficiency:</i> Hair loss, delayed wound healing, rashes, anorexia, hypogonadism in males, and difficulty with taste and smell <i>Excess:</i> Abdominal cramping and diarrhea

*Denotes primary source of specific nutrient.

Water

Next to oxygen, water is the most critical component to sustain life. Every cell, tissue, organ, and body system depends on adequate water intake to function properly. Water actually comprises more than half of the body's weight: 55% to 60% in adult females, 60% to 65% in adult males, and 65% to 70% in full-term infants. [Chapter 29](#) provides more information about the percentages of water found in individuals of different ages, in addition to the water composition of the brain, lungs, and blood. Refer to [Box 23.6](#) for a list of the functions of water in the body.

The Institute of Medicine recommends that the average male adult drink 14 to 15 eight-ounce glasses of fluid per day and the average female adult approximately 11 glasses per day. There are other researchers who believe the average adult only needs a minimum of six to eight 8-ounce glasses, or 1,440 to 1,920 mL, of water per day. Amid the controversy surrounding the minimum volume of water and fluid required for optimal health, most tend to follow the lesser recommendations of eight 8-ounce glasses per day. We recommend following the latter guideline of six to eight glasses per day.

The volume required will also vary according to the level of activity, health status, age, size, and degree of body fat. Exercise and physical exertion will increase the body's water requirement because of the losses through increased metabolism and perspiration. When the environmental temperatures are hot, as in the summer, or when a person is working in a hot environment, their water requirements will naturally rise. An individual with congestive heart failure or renal failure may be more restricted in fluid volume intake. Someone with a bacterial or viral infection will need to drink more fluids to help flush the toxins from the body. Fever, vomiting, and diarrhea deplete fluid volumes, thus increasing the need for additional fluids. Those individuals who have a higher body fat content may require more fluids than lean individuals, as fat holds very little water compared with lean muscle. It is particularly pertinent that pregnant or nursing women drink more fluids than nonpregnant women.

KNOWLEDGE CONNECTION

Explain at least four functions of water. What is the commonly recommended volume of fluid intake per day? Which groups of people require more intake of water than the average person? How much water do you drink daily?

Box 23.6

Functions of Water

- Maintains body temperature
- Transports minerals, vitamins, and electrolytes
- Cushions the brain and spinal cord
- Lubricates joints
- Carries toxins and excreted waste products out of the body
- Moisturizes and lubricates the gastrointestinal tract to aid the passage of food through the intestines

Real-World Connection

If a Little Is Good, Is More Better?

A young registered nurse who had only been out of nursing school for 3 years was very health conscious and physically active. She exercised 5 to 6 days a week, participated in various sports, drank lots of water, took time to have fun and play regularly, worked to eat a balanced diet, and took a daily multiple vitamin. After having several bouts of colds and allergies, she decided to boost her immunity by taking a vitamin C supplement, adding up to 4,000 mg of vitamin C per day. After 7 months she got tired of taking so many pills every day and discontinued the supplement. A week or so later she noticed large areas of *petechiae*, pinpoint-size flat red spots caused by capillary ruptures, over much of her body. The rash continued to spread for 4 days. Although the rash did not itch, it was a source of concern, and she decided to seek medical attention. Ultimately, it was determined that even though she ate well and was in good nutritional condition, she was suffering from some of the symptoms of scurvy, a disease caused by inadequate intake of vitamin C. By ingesting such high levels of vitamin C supplements for several months and then stopping the supplements cold turkey rather than weaning herself off them slowly, she had subjected her body to a *faux* scurvy. Even though she was still getting adequate vitamin C in her diet, her body misinterpreted the sudden and dramatic drop in vitamin C intake to be a deficiency of vitamin C.

DIFFERING NUTRITIONAL NEEDS OVER THE LIFE SPAN

Nutritional requirements constantly change throughout one's life span. Certain factors such as age, sex, periods of high growth rate, activity level, illness and injuries, and pregnancy all will contribute to the changing needs of each individual.

Toddlers and Preschool-Age Children

After the age of 1 year, a child may be switched from breastfeeding or formula to cow's milk. Children need a serving of milk or another dairy product with each meal because their bones and teeth are growing rapidly. It is recommended that whole milk be used as opposed to reduced-fat milk because young children require a certain level of fat in their diets. Approximately 30% to 35% of caloric intake should come from fats because dietary fats are vital to the formation of healthy nerves and brain.

Small children need to consume a wide variety of foods to meet their nutrient requirements for the rapid growth that occurs during this time. To meet needs for iron and protein, offer a peanut butter sandwich on whole-grain bread. Approximately one-fourth of each meal should come from whole grains, which contain more nutrients than refined grains. Whole wheat pasta and bread help provide these requirements.

To encourage small children to eat a wide variety of vegetables, offer "finger foods" such as the following:

- Raisins
- Grapes
- Sliced apples
- Sliced bananas
- Cut-up carrots
- Cherry tomatoes
- Broccoli, cauliflower, and cucumbers, along with dip

Safety: For the very young child, it will be necessary to cut the fruits and veggies into smaller bite-size pieces to avoid choking. (See Table 23.6 for serving recommendations for children of this

bite-size pieces to avoid choking. (See [Table 23.6](#) for serving recommendations for children of this age.)

The American Academy of Pediatrics recommends that standard children’s vitamin supplements are only necessary for children with poor appetites, those with health conditions that affect their nutritional intake, or those who eat restricted diets, such as a vegetarian diet. **Safety: Mega doses of vitamins should never be given to children except under a pediatrician’s direct order and care. Because many children’s vitamins are made to take in the form of gummy bears or sweet treats, it is important to teach parents to keep all vitamins out of the child’s reach to prevent accidental overdose.**

School-Age Children

It is important to gauge the quantities of food to be consumed according to the activity level of the child to prevent the child from becoming overweight and obese. However, school-age children will experience growth spurts at times that will increase their energy needs and thus their caloric intake.

School-age children need protein and calcium at higher rates than younger children. It is important to help children develop healthy eating habits that will generally last a lifetime. Encourage children to snack on healthier choices, such as the following:

- Dried fruits, nuts, and seeds (trail mix)
- Whole fruits and berries
- Raw vegetables with dip

In addition, encourage children to drink water more frequently than sugary beverages. Water flavored with slices of fruit such as strawberries, oranges, or melon may help it be more appealing to the school-age child. (See [Table 23.6](#) for serving recommendations for children of this age.)

Adolescents

Adolescence brings additional periods of rapid growth, usually between 10 and 15 years of age for girls and between 12 and 14 years for boys. This is a period of great energy expenditure, which increases the need for nutrients, especially calcium, protein, iron, vitamins, and minerals.

Healthy eating patterns established in childhood are beneficial to adolescents. They may feel a great deal of pressure to be a certain body size or shape and to look a certain way. Such pressure may lead to anorexia nervosa or bulimia.

Teens are often very busy with many activities and are eating on the go. Fast foods are readily available but often lack the nutrition needed by this age group. Encourage foods that are high in calcium and other nutrients, such as the following:

- Milk, cheese sticks, and yogurt
- Green leafy vegetables and salads
- Calcium-fortified juices
- Cereals
- Soy products
- Enriched breads and bread products
- Vegetables and fruits

Some fast-food restaurants and convenience foods are making an effort to offer healthier food items. Encourage adolescents to make better choices and eat more fruits and vegetables as half of their meal. (See [Table 23.6](#) for serving recommendations for this age group.)

Table 23.6

Recommended Intake for Various Age Groups of Average Size and Activity Level						
	Fruit	Vegetables	Grains	Proteins	Dairy	Oils
Toddlers and Preschoolers	½–1 c	2/3–1 c	1 3/4–5 oz	2 oz	1 2/3–2 c	8–13 grams/day
School-age Children	1–2 c	1 1/2 –2 1/2 c	3–6 oz	2–5 1/2 oz	2–2 1/2 c	15–24 grams/day

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Children						grams/day
Adolescents	1 1/2-2 1/2 c	2 1/4 -4 c	6-10 oz	5-7 oz	3 c	24-57 grams/day
Adults	1 1/2-2 1/2 c	2-4 c	5-10 oz	5-7 oz	3 c	22-44 grams/day
Older Adults	1 1/2-2 c	2-3 c	5-9 oz	5-6 oz	3 c	22-34 grams/day

Patient Teaching Connection

Pregnancy: Weight Gain and Diet

You are working with a group of pregnant teenagers at a local clinic. Many of the teens express concern about weight gain during pregnancy. **Safety: It is essential to teach pregnant teens that weight gain and proper nutrition during pregnancy are important determinants of growth and proper development of the baby and that poor weight gain and abnormal development are directly associated with low calorie intake and inadequate intake of vital nutrients.** This makes it pertinent that the young mothers eat adequate amounts of the essential nutrients in an effort to prevent low birth weights and unhealthy babies. Explain that weight gain is based on a person's prepregnancy body mass index (BMI). Weight gain for a pregnant adolescent should total between 25 and 40 pounds (11.3 to 18.1 kg). During the second and third trimesters, adolescents should gain approximately 1 pound (2.2 kg) per week. Additional nutrient requirements, such as folic acid, iron, and vitamin C, should be emphasized. Explain to the teens that adequate calcium is also needed for healthy growth. The best way for a pregnant teenager to meet her dietary needs is to maintain a regular three-meal schedule with three nutritious snacks, keeping in mind that adequate servings from all five of the different food groups are important. A varied, balanced eating plan is the only way to provide enough nutrients and food energy during pregnancy and to allow for healthy development of the baby. Stress the importance of taking prenatal vitamins as directed throughout the pregnancy and lactation.

Adulthood

The nutritional needs of adults, as in every other age group, will vary somewhat depending on age, sex, height, weight, and level of activity, but they generally tend to decrease with age.

At the onset of adulthood, energy requirements usually reach a plateau that will last until one's mid-40s, after which they begin to decline, primarily because activity and the amount of muscle versus body fat declines. So as activity level begins to slow, the total calories consumed each day should correspondingly decrease. To assist with weight maintenance and optimal adult health, a regular exercise program is considered vital. Weight gain in adulthood is closely associated with the development of cardiovascular disease and diabetes.

While total calories required by the adult decreases, there are some nutrient requirements for women that increase with age, and numerous nutrient needs increase with pregnancy and lactation. As women age and get closer to menopause, with a decrease in the occurrence of pregnancy, their need for iron decreases. However, there is a need to increase calcium intake because of the decreased calcium absorption and the increased risk for osteoporosis in postmenopausal females. (See [Table 23.6](#) for serving recommendations for adults.)

Older Adults

Many factors, including decreased appetite, impaired digestion, decreased GI absorption and motility, decreased senses of smell and taste, less physical activity, and slower metabolism, affect the nutritional status of older adults. Bone and muscle mass decrease as a natural part of aging and becoming less active. More calories are stored as fat because of reduced energy needs.

As more fat is stored (commonly around the abdomen), more fatty acids are released into the bloodstream, raising lipid and glucose levels. Then insulin levels automatically rise in an effort to combat the elevated glucose levels in the blood, making the older adult at risk for development of diabetes and cardiovascular disease.

Weight-bearing exercise is necessary for calcium uptake into the bones, but this usually decreases with aging. Lack of weight-bearing exercise and menopause in women can lead to osteoporosis.

Sometimes when older adults are on a fixed income, they may find it difficult to afford the foods needed for healthy eating. However, this age group should be encouraged to eat different forms of

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protein and calcium if meat and dairy are too expensive. Older adults who live alone may not enjoy preparing food for one person and may dislike eating alone. Meals served to older adults at various locations in the community can help mitigate this nutritional issue.

Older adults need the same nutrients as they needed as a younger adult, but they may have increased needs for calcium and vitamin D for the absorption of calcium; increased fiber to combat the slowed GI motility that can cause constipation; adequate iron to prevent anemia; and adequate protein intake to help fight infection, restore muscle mass that is lost because of immobility, and to promote healing of fractures and wounds. (See [Table 23.6](#) for serving recommendations for older adults.)

KNOWLEDGE CONNECTION

Describe at least two differences in nutritional needs for each age group through the life span. According to these recommendations, what might be lacking in your daily diet?

Obesity

Being overweight or obese results from eating more calories than you use each day. This affects all ages over the life span. The calories that are not burned are stored and add body weight. Obesity is measured by many scales, most commonly through measurement of the **body mass index (BMI)**. This measurement uses a formula, along with an individual's weight and height, to calculate whether her or his level of fatness is within normal weight boundaries or whether she or he is underweight, overweight, or obese. The BMI is a useful tool when evaluating a person's health risks. Individuals whose BMI falls under 18.5 are generally considered to be underweight, and those with a BMI between 18.5 and 24.9 are within the normal weight range. Those whose BMI is between 25 and 29.9 are considered to be overweight, and those with a BMI of 30 or greater are classified as obese.

However, athletes have more muscle mass weight than a nonathletic person, and pregnant women weigh more because of the pregnancy than they otherwise weigh. These examples show that the BMI number may be inaccurate. However, the BMI is still the best tool to use as an indicator of obesity in the general population. The BMI calculator tool may be found at www.nhlbi.nih.gov/health/educational/lose_wt/BMI/bmicalc.htm. It is simple to use, and we recommend that you go to the Web site and play with the tool to get a better handle on the BMI.

KNOWLEDGE CONNECTION

What percentage of one's total daily caloric intake should come from fats? In which groups of people might the BMI calculation be inaccurate? Calculate your BMI. Do you need to modify your eating habits and weight?

INFLUENCES ON NUTRITION

When discussing nutrition with your patients, take the time to ask about food restrictions. Many patients restrict their diets because of food preferences, intolerances, allergies, and cultural and religious influences. Providing healthy, nutritious meals that align with a patient's values and beliefs is an important part of caregiving.

CRITICAL THINKING CONNECTION: POST CONFERENCE

At postconference, you talk about your patient and his misunderstandings about diet and how it affects constipation. He told you that he cannot eat fiber because it gives him gas. After you asked a few questions, you learned that he had tried only one time to increase the fiber in his diet, increasing his fiber intake from 14 to 15 g one day to 30 g the next day, rather than gradually increasing fiber intake by a couple of grams per day. You know that such a large increase all at one time could cause flatus, and you were able to explain to him how to gradually increase fiber in his diet to prevent that from happening. You also helped him choose foods he likes and can easily prepare that have higher fiber content than what he has been eating. In addition, you were able to explain to him how drinking tea, coffee, or water would help increase the amount of fluid his body could use, while alcohol can cause the body to lose water.

the amount of fluid his body could use, while colas can cause the body to lose water.

Key Points

- The *Dietary Guidelines for Americans 2020–2025* encourage more variety of vegetable and protein sources, in addition to increased intake for whole fruits and whole grains, while limiting added sugars, saturated fats, and sodium.
- Essential nutrients include macronutrients (protein, carbohydrates, fats, and water) and micronutrients (vitamins and minerals).
- Protein is the primary building block of the body to make new cells and help wounds heal. People must consume foods with all nine essential amino acids (called complete proteins) or a combination of foods that provide them.
- There are many types of dietary fats, including monounsaturated, polyunsaturated, saturated, and trans fats. Dietary fats are needed as a concentrated source of energy, to help absorb and store fat-soluble vitamins, and to support and insulate internal organs.
- Carbohydrates serve as the primary source for energy and are more quickly digested into a usable form than either proteins or fats. Carbohydrates may be simple or complex. Fiber is an indigestible carbohydrate and is needed by the body.
- Fat-soluble vitamins are stored in body fat and include vitamins A, D, E, and K. Water-soluble vitamins are not stored and must be consumed daily. They include all of the B vitamins and vitamin C.
- Sources of vitamins vary greatly. The body cannot manufacture them, so they must be consumed in the daily diet and by exposure to the sun (vitamin D). They are needed for a variety of metabolic processes.
- Minerals are inorganic compounds needed in small amounts by the body. Major minerals are those we need to ingest 100 mg per day for optimal health. Trace minerals are those we require less of each day.
- Water is a critical component to sustain life and is necessary for the whole body to function correctly.
- Different age groups need differing amounts of the food groups: vegetables, fruits, grain, protein, dairy, and oils. However, all age groups need all food groups. Healthy eating choices should be taught from an early age to help form a lifetime of better eating.
- Obesity is defined as a BMI of 30 or greater. Overweight is defined by a BMI between 25 and 29.9.

CRITICAL THINKING EXERCISES

Answers available online.

1. Mr. Baze is 71 years old and is being seen in the clinic for hypertension. His primary care provider wants him to follow a low-sodium diet. He asks you about foods that he should avoid. What is your response?
2. Ms. Havner is concerned about her nutritional status. She has access to the Internet and has become confused about the many nutritional Web sites. What information can you give her to assess her current nutritional status?

Additional Resources



Go to Davis Advantage to complete your learning: strengthen understanding, apply your knowledge, and prepare for the Next Gen NCLEX®.



Study Guide

